4.4.5 Activators

Activators are the inorganic ions (cations or anions) or compounds that enhances the activity of the enzymes.

Enzyme	Activator
Phenol oxidase	Cu^{2+}
Amylase	Cl^{-}

Metal ions can be bound to the enzyme permanently or can be used just for activation. When metal ions used only for the activation of enzyme, they are called metal activated enzymes.

Examples: ATPase (Mg^{2+}, Ca^{2+}) and Enolase (Mg^{2+}).

If metal ions are bound with enzymes using chemical bond, they are called metalloenzymes.

Examples: Alcohol dehydrogenase-Zn^{2+} and Carbonic anhydrase-Zn^{2+},

Time, radiations and co-enzymes are also the other factors which affect the velocity of an enzyme reaction.

4.5 Inhibitors

An inhibitor is defined as a substance which binds with the enzyme and brings about a decrease in catalytic activity of that enzyme. For example, anti-oxidants are added as inhibitors to food to retard its spoilage on exposure to air (oxygen) and inhibition could be either reversible or irreversible.

Allosteric activators and inhibitors:

This type of inhibition takes place due to the presence of allosteric site (Greek allo = 'other'; stereos = 'space' or 'site') on the surface of the enzyme away from the active site. The final end-product fits in the allosteric site and in some way brings about a change in shape of the enzyme so that the active site of the enzyme becomes unfit for making a complex with its substrate. Allosteric inhibition may be reversible. In many metabolic reactions, when the concentration of the final end product (usually acts as an allosteric inhibitor) in the cell falls and the activity of the enzyme is restored. Similary an enzyme can also be activated by an activator that binds to an allosteric site. This activator is called as an allosteric activator.

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